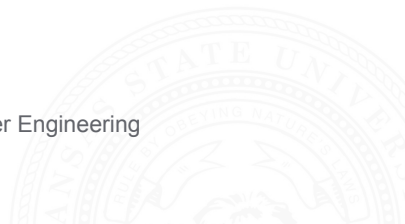


Three-Phase AC Circuits II

Three-phase power, per-phase equivalents, and
complete feeders

Buxin She

Mike Wiegiers Department of Electrical and Computer Engineering



Lecture outline

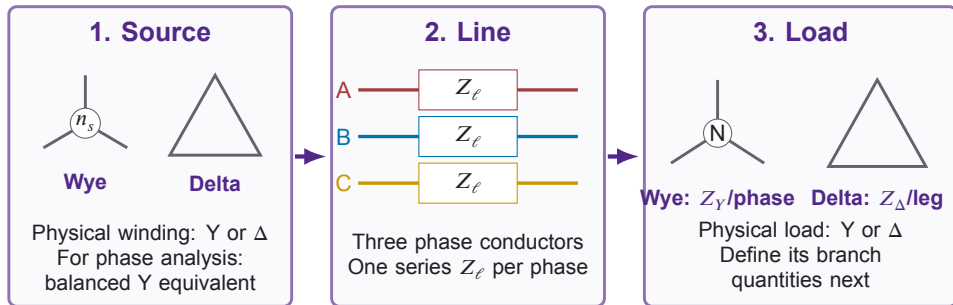
- 1 Per-phase analysis
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- 3 Wye feeder example
- 4 Delta feeder example

PART 1

Per-phase analysis

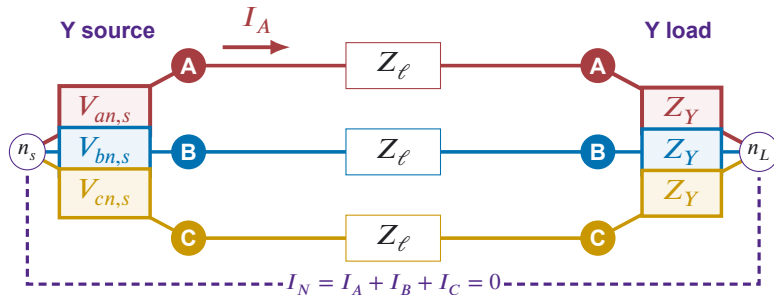
Balanced circuits and delta-to-wye equivalence

Identify the source, line, and load first



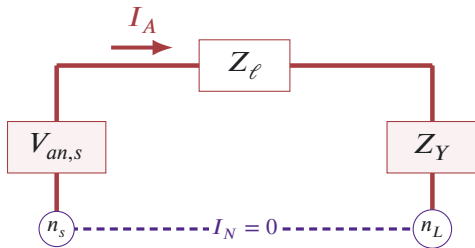
The source representation, line model, and physical load connection are three separate modeling choices.

Start with a balanced Y–Y network



Balanced Y–Y has three identical phase paths; solve one phase and rotate it by $\pm 120^\circ$.

Extract one phase, then reconstruct three



Why one phase is enough

The balanced Y–Y circuit has three identical phase paths. Solving the A-phase loop therefore determines the other two by rotation.

Solve the highlighted loop

$$I_A = \frac{V_{an,s}}{Z_\ell + Z_Y}.$$

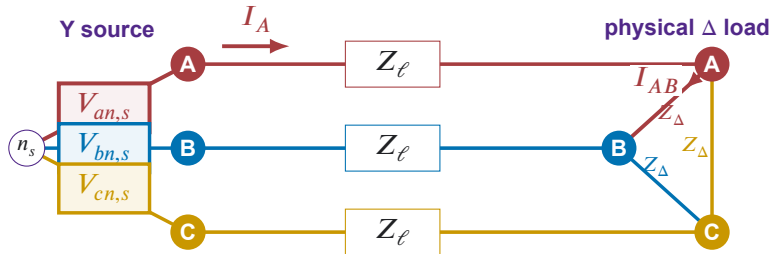
Recover the balanced set

$$I_B = I_A \angle (-120^\circ),$$

$$I_C = I_A \angle (+120^\circ).$$

For a Y load, solve one phase and rotate; next ask what changes when the load is Δ .

A Y source feeding a Δ load

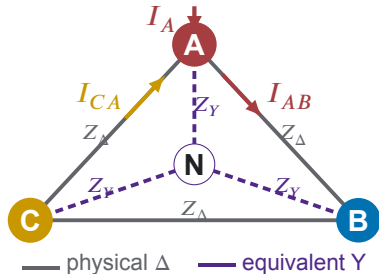


Now replace only the Y load by a physical Δ load. Every load branch lies between two line terminals, so the load neutral and the direct A–N phase path used on the preceding page no longer exist.

To reuse the one-phase method, replace the physical Δ load by a terminal-equivalent Y.

Compare Y and Δ at the same terminals

Same A–B–C boundary; two alternatives



Use the same terminal voltages

$$V_{AB} = V_{AN} - V_{BN}, \quad V_{CA} = V_{CN} - V_{AN}.$$

Read the currents from the diagram

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}}, \quad I_{CA} = \frac{V_{CA}}{Z_{\Delta}},$$
$$I_A^{[\Delta]} = I_{AB} - I_{CA}, \quad I_A^{[Y]} = \frac{V_{AN}}{Z_Y}.$$

N is the common voltage reference and the Y neutral; the Δ has no physical neutral. Also, I_{AB} is a leg current, not the line current.

$$I_A^{[\Delta]} = I_A^{[Y]}.$$

Derive the balanced impedance relation

Delta terminal current

$$\begin{aligned} I_A^{[\Delta]} &= I_{AB} - I_{CA} \\ &= \frac{V_{AB} - V_{CA}}{Z_{\Delta}} \\ &= \frac{(V_{AN} - V_{BN}) - (V_{CN} - V_{AN})}{Z_{\Delta}} \\ &= \frac{2V_{AN} - V_{BN} - V_{CN}}{Z_{\Delta}} \\ &= \frac{3V_{AN}}{Z_{\Delta}}. \end{aligned}$$

The last step uses $V_{AN} + V_{BN} + V_{CN} = 0$.

Wye terminal current

$$I_A^{[Y]} = \frac{V_{AN}}{Z_Y}.$$

Match the A-terminal behavior

$$I_A^{[Y]} = I_A^{[\Delta]} \implies \frac{V_{AN}}{Z_Y} = \frac{3V_{AN}}{Z_{\Delta}}.$$

$$\boxed{Z_Y = \frac{Z_{\Delta}}{3}},$$

$$\boxed{Z_{\Delta} = 3Z_Y}.$$

Use the balanced $\Delta \leftrightarrow Y$ conversion

When the simple rule applies

- three identical impedances;
- the same terminals A, B, and C;
- balanced terminal voltages.

Impedance conversion

$$\Delta \rightarrow Y : \quad Z_Y = \frac{Z_{\Delta}}{3},$$

$$Y \rightarrow \Delta : \quad Z_{\Delta} = 3Z_Y.$$



What the equivalent preserves

- terminal voltages and line currents;
- total three-phase complex power.

What must be recovered

- the physical connection;
- delta branch voltages and currents;
- any internal neutral or branch quantity.

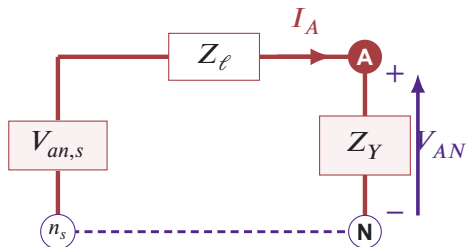
PART 2

Three-phase power

Complex power in balanced circuits

Start from the Y–Y diagram: one-phase power

Load branch variables



Passive load reference

All phasor magnitudes are RMS.

$$V_{AN} = V_\phi e^{j\theta_v}, \quad I_A^* = I_\phi e^{-j\theta_i}.$$

$$S_A = V_{AN} I_A^* = V_\phi I_\phi e^{j(\theta_v - \theta_i)}.$$

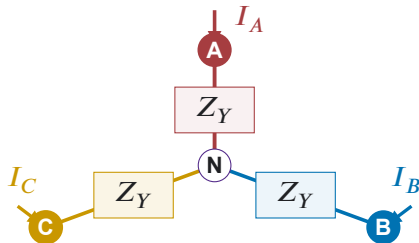
$$P_A = V_\phi I_\phi \cos \phi,$$

$$Q_A = V_\phi I_\phi \sin \phi, \quad \phi = \theta_v - \theta_i.$$

Matched branch power: $S_A = V_{AN} I_A^*$, with I_A entering the positive-voltage terminal.

Sum the three Y-connected load branches

One matched pair per branch



Branch pairs: (V_{AN}, I_A) , (V_{BN}, I_B) ,
and (V_{CN}, I_C) .

The rotations cancel in each VI^*

$$V_{BN} = V_{AN}e^{-j120^\circ}, \quad I_B^* = I_A^*e^{j120^\circ},$$

$$V_{CN} = V_{AN}e^{j120^\circ}, \quad I_C^* = I_A^*e^{-j120^\circ}.$$

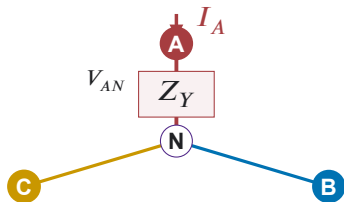
$$S_B = V_{BN}I_B^* = S_A, \quad S_C = V_{CN}I_C^* = S_A.$$

$$\boxed{S_{3\phi} = 3V_{AN}I_A^*}, \quad S_{3\phi} = P_{3\phi} + jQ_{3\phi}.$$

The Y diagram supplies three matched phase products; balance makes them equal.

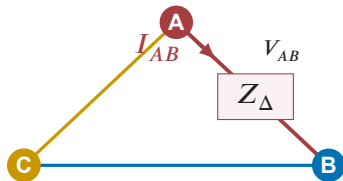
Change the load connection: Y versus Δ

Y load: use one phase-to-neutral branch



$$S_{3\phi} = 3V_{AN}I_A^* = 3|I_A|^2 Z_Y.$$

Δ load: use one physical delta leg



$$S_{3\phi} = 3V_{AB}I_{AB}^* = 3|I_{AB}|^2 Z_\Delta.$$

Matched pairs: Y uses (V_{AN}, I_A) ; Δ uses (V_{AB}, I_{AB}) .

Derive both phase and line forms

Start from the matched physical branch on each diagram, then convert its RMS magnitudes to line quantities.

Y connection

$$V_\phi = V_{LL}/\sqrt{3}, \quad I_\phi = I_L$$



$$S_{3\phi} = 3V_{AN}I_A^*,$$

$$|S_{3\phi}| = 3V_\phi I_\phi = 3 \frac{V_{LL}}{\sqrt{3}} I_L = \sqrt{3} V_{LL} I_L.$$

Δ connection

$$V_\phi = V_{LL}, \quad I_\phi = I_L/\sqrt{3}$$



$$S_{3\phi} = 3V_{AB}I_{AB}^*,$$

$$|S_{3\phi}| = 3V_\phi I_\phi = 3V_{LL} \frac{I_L}{\sqrt{3}} = \sqrt{3} V_{LL} I_L.$$

Use matched phase/branch phasors for complex power; convert their magnitudes only when using the line form.

Three-phase power: phase and line forms

$S_{3\phi} = P_{3\phi} + jQ_{3\phi}$ is a complex scalar; P , Q , and $|S|$ are real scalars. Voltage and current entries below are RMS magnitudes, and ϕ is the matched-current lag angle.

Quantity	Matched phase/branch form	Line-quantity form
$S_{3\phi}$	$3V_{\phi}I_{\phi}e^{j\phi}$	$\sqrt{3}V_{LL}I_L e^{j\phi}$
$ S_{3\phi} $	$3V_{\phi}I_{\phi}$	$\sqrt{3}V_{LL}I_L$
$P_{3\phi}$	$3V_{\phi}I_{\phi} \cos \phi$	$\sqrt{3}V_{LL}I_L \cos \phi$
$Q_{3\phi}$	$3V_{\phi}I_{\phi} \sin \phi$	$\sqrt{3}V_{LL}I_L \sin \phi$

$$(V_{\phi}, I_{\phi}) = \begin{cases} (V_{AN}, I_A), & \text{Y branch,} \\ (V_{AB}, I_{AB}), & \Delta \text{ branch.} \end{cases} \quad \text{pf} = \cos \phi.$$

Inductive: $\phi > 0$, $Q > 0$, lagging

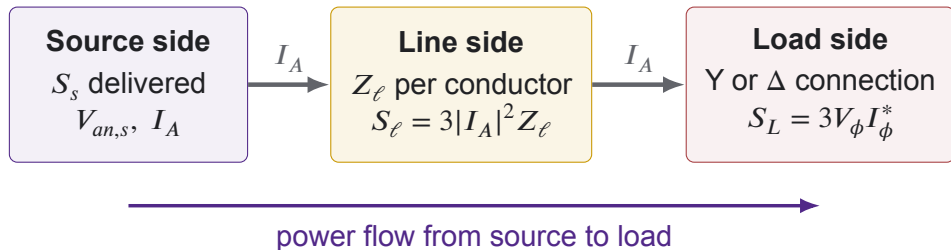
Resistive: $\phi = 0$, $Q = 0$, unity

Capacitive: $\phi < 0$, $Q < 0$, leading

The phase/branch and line forms describe the same balanced three-phase power at the same circuit boundary.

From load power to feeder power balance

The preceding formulas evaluated power at the load terminals. A complete feeder also absorbs power in its three line impedances.



$$\boxed{S_s = S_\ell + S_L}, \quad P_s = P_\ell + P_L, \quad Q_s = Q_\ell + Q_L.$$

The earlier formulas give S_L ; add the line absorption before comparing source delivery with load absorption.

Why line power uses $|I_A|^2$

I_A is the phase-A RMS current phasor; $I_L \equiv |I_A|$ is its real scalar magnitude.

Start from one line impedance

The current angle cancels

$$\begin{aligned}V_{\ell,A} &= Z_{\ell} I_A, \\S_{\ell,A} &= V_{\ell,A} I_A^* \\&= (Z_{\ell} I_A) I_A^* \\&= Z_{\ell} (I_A I_A^*) = |I_A|^2 Z_{\ell}.\end{aligned}$$

$$\begin{aligned}I_A &= I_L \angle \theta_I, \\I_A^* &= I_L \angle (-\theta_I), \\I_A I_A^* &= I_L^2 \angle 0^\circ \\&= |I_A|^2.\end{aligned}$$

RMS convention

If $\hat{I}_A$ is a peak magnitude, then $|I_A|^2 = \hat{I}_A^2/2$.

$$\boxed{S_{\ell} = 3|I_A|^2 Z_{\ell}}, \quad P_{\ell} = 3|I_A|^2 R_{\ell}, \quad Q_{\ell} = 3|I_A|^2 X_{\ell}.$$

For line absorption, the current angle cancels because $V_{\ell,A} = Z_{\ell} I_A$; therefore $\angle S_{\ell} = \angle Z_{\ell}$.

Source, line, and load power balance

$S = P + jQ$ is a complex scalar; P and Q are real scalars. Source voltage and current are independent phasors, whereas line voltage satisfies $V_{\ell,A} = Z_{\ell} I_A$.

	Complex power	Power angle
Source	$S_s = 3V_{an,s} I_A^*$	$\angle V_{an,s} - \angle I_A$
Line	$S_{\ell} = 3 I_A ^2 Z_{\ell}$	$\angle Z_{\ell}$

Apply the phase KVL equation

$$V_{an,s} = I_A Z_{\ell} + V_{AN,L},$$

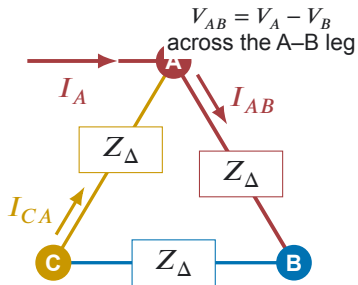
$$3V_{an,s} I_A^* = 3(I_A Z_{\ell}) I_A^* + 3V_{AN,L} I_A^*.$$

$$\boxed{S_s = S_{\ell} + S_L}, \quad \boxed{P_s = P_L + 3|I_A|^2 R_{\ell}}, \quad \boxed{Q_s = Q_L + 3|I_A|^2 X_{\ell}}.$$

The complex balance $S_s = S_{\ell} + S_L$ yields separate real-scalar balances for P and Q .

Example: complex power of one delta leg

Balanced positive-sequence Δ load



Given branch quantities

$$V_{AB} = 13.8\angle 0^\circ \text{ kV},$$
$$I_{AB} = 0.138\angle (-20^\circ) \text{ kA}.$$

$$\begin{aligned} S_{AB} &= V_{AB} I_{AB}^* \\ &= (13.8\angle 0^\circ)(0.138\angle 20^\circ) \\ &= 1.9044\angle 20^\circ \text{ MVA} \\ &= 1.78955 + j0.65134 \text{ MVA}. \end{aligned}$$

Specify the Δ connection and the A $\rightarrow$ B branch direction before using $S_{AB} = V_{AB} I_{AB}^*$.

Example: power of the 13.8-kV delta

$$I_A = I_{AB} - I_{CA} = \sqrt{3}I_{AB}\angle(-30^\circ) \\ = 239.023\angle(-50^\circ) \text{ A},$$

$$I_{AB} : A \rightarrow B, \quad I_{CA} : C \rightarrow A.$$

$$S_{3\phi} = 3V_{AB}I_{AB}^* = 3(13.8)(0.138)\angle 20^\circ = 5.7132\angle 20^\circ \text{ MVA}.$$

Power components

$$S_{3\phi} = 5.36865 + j1.95403 \text{ MVA}$$

$$P = 5.36865 \text{ MW}, Q = 1.95403 \text{ Mvar};$$

$$\text{pf} = \cos 20^\circ = 0.9397 \text{ lagging}.$$

I_{AB} is a Δ -leg current and I_A is the line current; use the matched pair V_{AB} , I_{AB} for complex power.

Independent magnitude check

$$\sqrt{3}(13.8 \text{ kV})(0.239023 \text{ kA})$$

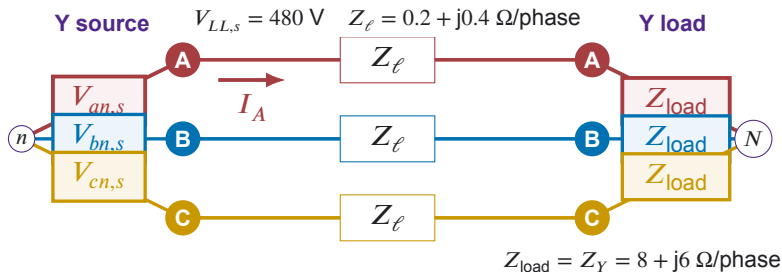
$$= 5.7132 \text{ MVA}.$$

PART 3

Wye feeder example

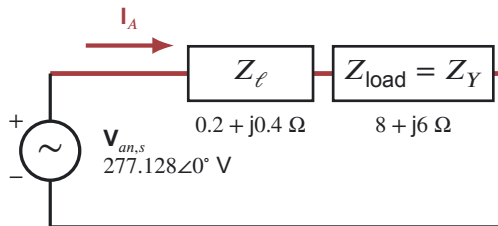
Voltage drop and power balance in a 480-V feeder

Example A: 480-V wye feeder



Equal source magnitudes, line impedances, and phase loads preserve balance and permit phase-A analysis.

Example A: per-phase circuit



$$Z_{\text{eq}} = Z_\ell + Z_Y = 8.2 + j6.4 \Omega$$

$$Z_{\text{eq}} = 10.402\angle 37.972^\circ \Omega.$$

The other phases have identical circuits rotated by $\mp 120^\circ$.

Example A: solve the phase current

We choose $V_{an,s} = 277.128\angle 0^\circ$ V; hence $V_{ab,s} = 480\angle 30^\circ$ V.

$$I_A = \frac{480/\sqrt{3}}{(0.2 + j0.4) + (8 + j6)} = \boxed{26.642\angle(-37.972^\circ) \text{ A}} = 21.002 - j16.392 \text{ A}.$$

Recover the balanced current set

$$I_B = 26.642\angle(-157.972^\circ) \text{ A},$$

$$I_C = 26.642\angle 82.028^\circ \text{ A}.$$

Interpret the current

Wye: $I_L = I_\phi = 26.642$ A.

Current lags source phase voltage
because $\Im(Z_\ell + Z_Y) > 0$.

Use the original input values in subsequent calculations; displayed results are rounded.

Example A: line voltage drop

Use the solved $I_A = 21.0023166 - j16.3920520$ A and $Z_\ell = 0.2 + j0.4 \Omega$:

$$\begin{aligned}\Delta V_A &= I_A Z_\ell \\ &= (21.0023166 - j16.3920520)(0.2 + j0.4) \\ &= (4.2004633 + 6.5568208) + j(8.4009267 - 3.2784104) \\ &= \boxed{10.757 + j5.123 \text{ V}}.\end{aligned}$$

$$\Delta V_A = 11.915 \angle 25.463^\circ \text{ V}.$$

The drop angle is the current angle plus the line-impedance angle. The drop is a complex voltage.

Example A: load phase voltage and KVL

$$\begin{aligned} V_{AN,L} &= I_A Z_Y = (21.0023166 - j16.3920520)(8 + j6) \\ &= 266.371 - j5.123 = 266.420 \angle (-1.102^\circ) \text{ V.} \end{aligned}$$

Independent source-minus-drop route

$$V_{an,s} - \Delta V_A = 277.128 - (10.757 + j5.123) = 266.371 - j5.123 \text{ V.}$$

$$\boxed{V_{AN,L} + \Delta V_A = 277.128 + j0 = V_{an,s} .}$$

The load-product route and the source-minus-drop route give the same phase voltage.

Example A: load line-to-line voltage

From $V_{AN,L} = 266.420\angle(-1.102^\circ)$ V:

$$V_{AB,L} = \sqrt{3}V_{AN,L}\angle 30^\circ = \boxed{461.453\angle 28.898^\circ \text{ V}}.$$

Check by subtracting load phase voltages

$$V_{BN,L} = 266.420\angle(-121.102^\circ) = -137.622 - j228.123 \text{ V}.$$

$$V_{AN,L} - V_{BN,L} = 403.992 + j223.000 \text{ V} = 461.453\angle 28.898^\circ \text{ V}.$$

The load line voltage is reconstructed at the load terminals. Its magnitude is below the 480-V source rating in this case.

Example A: load complex power

The balanced wye load has $Z_Y = 8 + j6 \, \Omega$ and $|I_A| = 26.64200955 \, \text{A}$.

$$\begin{aligned} S_L &= 3V_{AN,L}I_A^* = 3|I_A|^2 Z_Y \\ &= 3(26.64200955)^2(8 + j6) \\ &= \boxed{17.035 + j12.776 \, \text{kVA}}. \end{aligned}$$

Apparent-power magnitude

$$|S_L| = 21.294 \, \text{kVA}.$$

Load power factor

$$\text{pf}_L = 8/\sqrt{8^2 + 6^2} = 0.800 \text{ lagging}.$$

The load impedance sets the load power angle. The source also supplies the line absorption.

Example A: complete power balance

Three-phase quantity	Calculation	P (kW)	Q (kvar)
Source delivery	$3V_{an,s}I_A^*$	17.461	13.628
Line absorption	$3 I_A ^2Z_\ell$	0.426	0.852
Load absorption	$3 I_A ^2Z_Y$	17.035	12.776

$$S_s = S_\ell + S_L = (0.426 + j0.852) + (17.035 + j12.776) \text{ kVA.}$$

$$|S_L| = 21.294 \text{ kVA,} \quad \text{pf}_L = \cos(\angle Z_Y) = 0.800 \text{ lagging.}$$

Both the line and the load absorb reactive power. The source supplies their complex-power sum.

Example A: independent checks

- 1 $I_A + I_B + I_C = 0$ from the reconstructed positive-sequence set.
- 2 $V_{an,s} - V_{AN,L} = I_A Z_\ell$ in rectangular form.
- 3 $S_L = 3V_{AN,L} I_A^* = 3|I_A|^2 Z_Y = 17.035 + j12.776 \text{ kVA}$.
- 4 $|S_L| = \sqrt{3}(461.453 \text{ V})(26.642 \text{ A}) = 21.294 \text{ kVA}$.

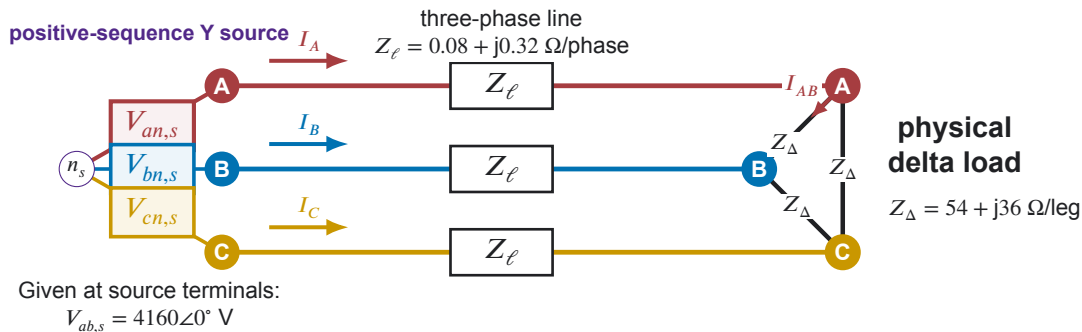
The line-magnitude check must use load voltage for load power, not the 480-V source rating.

PART 4

Delta feeder example

Equivalent-wye analysis of a 4.16-kV delta load

Example B: 4.16-kV delta feeder



Calculate I_A , load V_{AB} , physical I_{AB} , and source–line–load complex powers; then verify KVL and power balance.

Example B: delta-to-wye load conversion

The physical load has three equal delta legs, $Z_{\Delta} = 54 + j36 \Omega$.

$$Z_Y = \frac{Z_{\Delta}}{3} = \frac{54 + j36}{3} = \boxed{18 + j12 \Omega}.$$

$$|Z_Y| = \sqrt{18^2 + 12^2} = 21.633 \Omega, \quad \angle Z_Y = 33.690^\circ.$$

Preserved terminal quantities

Line currents, terminal line voltages, and total load power.

Quantities to recover later

The original delta-leg current uses the physical $Z_{\Delta} = 54 + j36 \Omega$.

The current in the equivalent wye branch is the physical line current.

Example B: source reference and phase circuit

The given source phasor is $V_{ab,s} = 4160\angle 0^\circ \text{ V}$.

$$V_{an,s} = \frac{4160}{\sqrt{3}} \angle (-30^\circ) = \boxed{2401.777 \angle (-30^\circ) \text{ V}}.$$

$$Z_{\text{tot}} = Z_\ell + Z_Y = (0.08 + j0.32) + (18 + j12) = 18.08 + j12.32 \ \Omega.$$

$$Z_{\text{tot}} = 21.879 \angle 34.271^\circ \ \Omega.$$

The phase circuit uses the converted source angle, the actual line impedance, and the equivalent load impedance.

Example B: solve the line current

$$Z_{\text{tot}} = Z_{\ell} + Z_Y = 18.08 + j12.32 = 21.879 \angle 34.271^\circ \Omega,$$

$$I_A = \frac{V_{an,s}}{Z_{\text{tot}}} = \boxed{109.778 \angle (-64.271^\circ) \text{ A}}.$$

The current angle equals the source phase angle minus the total-impedance angle.

Example B: load voltage and KVL

$$V_{AN,L} = I_A Z_Y = \boxed{2374.860 \angle (-30.581^\circ) \text{ V}},$$
$$V_{an,s} - V_{AN,L} = 35.459 + j7.338 \text{ V} = I_A Z_\ell.$$

The vector voltage-drop equation closes before any line-to-line quantity is reconstructed.

Example B: load terminal line voltage

The equivalent load phase voltage is $V_{AN,L} = 2374.860\angle(-30.581^\circ)$ V.

$$\begin{aligned} V_{AB,L} &= \sqrt{3}V_{AN,L}\angle 30^\circ \\ &= \sqrt{3}(2374.860)\angle(-30.581^\circ + 30^\circ) \\ &= \boxed{4113.379\angle(-0.581^\circ) \text{ V}}. \end{aligned}$$

$$V_{BC,L} = 4113.379\angle(-120.581^\circ) \text{ V}, \quad V_{CA,L} = 4113.379\angle 119.419^\circ \text{ V}.$$

Each original delta leg sees one of these load terminal voltages. The source rating is 4160 V.

Example B: physical delta branch currents

Use the original $Z_{\Delta} = 54 + j36 = 64.900\angle 33.690^{\circ} \Omega$:

$$I_{AB} = \frac{4113.379\angle(-0.581^{\circ})}{64.900\angle 33.690^{\circ}} = \boxed{63.380\angle(-34.271^{\circ}) \text{ A}}.$$

$$I_{BC} = 63.380\angle(-154.271^{\circ}) \text{ A}, \quad I_{CA} = 63.380\angle 85.729^{\circ} \text{ A}.$$

Magnitude check

109.778/63.380 = 1.73205 using
unrounded values.

Angle check

$$-64.271^{\circ} - (-34.271^{\circ}) = -30^{\circ}.$$

The solved line current and recovered delta branch current satisfy the same connection relation derived in L08.

Example B: source complex power

$$V_{an,s} = 2401.777 \angle (-30^\circ) \text{ V}, \quad I_A = 109.778 \angle (-64.271^\circ) \text{ A}.$$

$$\begin{aligned} S_s &= 3V_{an,s}I_A^* \\ &= 3(2401.777)(109.778) \angle [-30^\circ - (-64.271^\circ)] \\ &= 790.987 \angle 34.271^\circ \text{ kVA} \\ &= \boxed{653.657 + j445.412 \text{ kVA}}. \end{aligned}$$

Source delivery uses the source phase voltage. Its power factor is $\cos 34.271^\circ = 0.8264$ lagging.

Example B: line and load absorption

Three line impedances

$$\begin{aligned} S_{\ell} &= 3|I_A|^2(0.08 + j0.32) \\ &= 2.892 + j11.569 \text{ kVA.} \end{aligned}$$

Independent calculation in the physical delta

Equivalent-wye load

$$\begin{aligned} S_L &= 3|I_A|^2(18 + j12) \\ &= 650.765 + j433.843 \text{ kVA.} \end{aligned}$$

$$\begin{aligned} S_L &= 3|I_{AB}|^2 Z_{\Delta} = 3(63.38033284)^2(54 + j36) \\ &= 650.765 + j433.843 \text{ kVA.} \end{aligned}$$

The equivalent wye and original delta produce the same total load power with their corresponding branch currents.

Example B: source, line, and load powers

Three-phase quantity	Calculation	P (kW)	Q (kvar)
Source delivery	$3V_{an,s}I_A^*$	653.657	445.412
Line absorption	$3 I_A ^2Z_\ell$	2.892	11.569
Load absorption	$3 I_A ^2Z_Y$	650.765	433.843

$$S_s = S_\ell + S_L, \quad S_L = 3V_{AB,L}I_{AB}^* = 3|I_{AB}|^2Z_\Delta.$$

Source

$|S_s| = 790.987$ kVA; $\text{pf}_s = 0.8264$ lagging.

Load

$|S_L| = 782.122$ kVA; $\text{pf}_L = 0.8321$ lagging.

Both delta and equivalent-wye power calculations agree. Line real loss is 0.442% of source P .

Example B: independent reconstruction checks

- 1 $I_B = 109.778\angle 175.729^\circ$ A and $I_C = 109.778\angle 55.729^\circ$ A give $I_A + I_B + I_C = 0$.
- 2 $V_{an,s} - V_{AN,L} = I_A Z_\ell = 35.459 + j7.338$ V.
- 3 $|I_A|/|I_{AB}| = 1.73205$ and $\angle I_A - \angle I_{AB} = -30.000^\circ$.
- 4 $\sqrt{3}V_{LL,s}I_L = 790.987$ kVA and $\sqrt{3}V_{LL,L}I_L = 782.122$ kVA agree with the corresponding complex-power magnitudes.

Restore the other phases by rotation, and check each power at the voltage of its own terminal location.

A complete balanced three-phase solution

- 1 Declare RMS, sequence, voltage endpoints, and current directions.
- 2 Form one phase: $V_{an} = V_{ab} \angle (-30^\circ) / \sqrt{3}$ and, if needed, $Z_Y = Z_\Delta / 3$.
- 3 Solve $I_A = V_{an,s} / (Z_\ell + Z_Y)$; recover load line voltage, other phases, and original delta branch currents.
- 4 Compute $S_{3\phi} = 3V_\phi I_\phi^*$ with matched RMS pairs; verify KVL, KCL, and $S_s = S_\ell + S_L$.

Use $\sqrt{3}V_{LL}I_L$ for apparent-power magnitude, and retain the correct phase reference for every complex-power product.